

Surendar Pala Danasekaran

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Full-stack developer with 2+ years of experience building scalable, accessible, and high-performance web and mobile applications across healthcare, enterprise SaaS, and fintech. Comfortable crafting modern frontends with React, Next.js, and Tailwind CSS, architecting end-to-end solutions with NestJS backends and GCP deployments, with a strong focus on secure multi-tenant systems, real-time features, and delivering products that teams and users rely on. Experienced leveraging AI coding tools (Claude Code, GitHub Copilot) to accelerate development.

EDUCATION

University of Ottawa <i>Master of Computer Science 9.71 CGPA</i>	Ottawa, Canada <i>Sep. 2024 – May. 2026</i>
SRM Institute of Technology and Science <i>BTech, Computer Science (IoT Specialization) 9.52 CGPA</i>	Chennai, India <i>Sep. 2020 – May 2024</i>

EXPERIENCE

Software Developer Co-op <i>University of Ottawa</i>	Sep. 2025 – Jan. 2026 <i>Ottawa, Canada</i>
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- Built **InvisGuard**, a radar-based patient monitoring PWA (Next.js, MQTT, Supabase, Prisma ORM) with push-notification support, deployed on GCP with Cloudflare CDN, delivering sub-second alert latency for 20+ concurrent patients.
- Engineered a secure multi-tenant backend with **fine-grained RBAC middleware** in Prisma, enforcing strict data isolation and encrypted MQTT device-to-cloud communication across multiple care organizations.
- Implemented a real-time device registration and configurable alert threshold engine with VAPID push notifications; established a test-driven development (TDD) approach using Jest.

Software Developer (Co-op to Contract, Part-time) <i>Orleans Cardiopulmonary Group</i>	May 2025 – Present <i>Orleans, Canada</i>
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- Developed **OCPG Health**, a React Native Expo mobile app for CHF patients with 60+ REST API endpoints, featuring JVP video assessments, AI-powered nutrition analysis (OpenAI), vitals tracking, medication management, and an educational quiz system.
- Built a doctor portal using Directus CMS and a scalable backend (NestJS, MySQL, Prisma ORM, Docker, GCP) with Better Auth and Google OAuth, integrating real-time notifications and streamlining clinical workflows for clinic staff.
- Engineered a secure JVP video pipeline using **RSA-SHA256 signed GCS URLs** for direct client-to-storage uploads and implemented a **streaming LLM proxy** to deliver low-latency AI nutrition analysis via Server-Sent Events alongside a gamified CHF University quiz system with achievement tracking to boost patient engagement.

Software Engineer (Part-time) <i>Stejassys</i>	Mar. 2024 – Present <i>Remote</i>
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- Built **Innkraft**, a multi-hotel conversational AI sales agent (NestJS, LangChain, LangGraph, OpenAI GPT) on GCP, featuring a multi-stage decoder sub-agent for structured intent extraction, a query router, and a response composer with Prisma/PostgreSQL-backed conversation state and live room availability via multiple booking providers.
- Architected **Plugginn**, a multi-tenant embeddable widget platform (Next.js, NestJS, Stripe, Sentry, database-backed rate limiting) serving 30+ business clients across 15+ sites, with automated CI/CD via GitHub Actions with Sentry-based observability, contributing to 99.9% uptime.
- Built a real-time ride-hailing monorepo (React Native Expo + NestJS microservices) tracking 3,000+ vehicles, leveraging **Redis** for low-latency location updates and rate limiting and validated app flows with **Maestro** for E2E test coverage.

Full Stack Software Developer (Contract) <i>Flookup Capital Advisors (One21.ai)</i>	Aug. 2022 – Mar. 2024 <i>Remote</i>
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- Architected and maintained a suite of 6+ applications for the One21.ai fintech ecosystem (Next.js, TypeScript, Astro, Vercel, AWS S3), including **CA GPT** (an LLM-powered assistant for accounting workflows) and **Merchant Banker**, streamlining financial data visualization and user onboarding.
- Engineered a centralized TypeScript backend repository providing shared REST APIs, strict data typings, and robust error handling across multiple client-facing frontends (Flookup, Snapshot, MCATech), accelerating delivery and reducing cross-repo regressions.
- Managed end-to-end Vercel CI/CD pipelines and modular frontend architectures; leveraged AI coding agents (Claude Code, GitHub Copilot) to accelerate feature delivery, write tests, and debug across the product suite, improving development velocity and code quality.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL (PostgreSQL, MySQL), HTML5, CSS3
Frontend & Mobile: React, Next.js, Astro, React Native (Expo), Tailwind CSS, Shadcn UI, Redux, Zustand, Recharts
Backend & Data: Node.js, NestJS, Express.js, REST APIs, GraphQL, Prisma ORM, Drizzle ORM, Redis (Caching/Pub-Sub), WebSockets, MQTT, LangChain, LangGraph
Cloud & DevOps: GCP (Cloud Run, GCS), AWS (S3), Docker, CI/CD (GitHub Actions), Cloudflare, Supabase, Firebase
Tools, Architecture & AI Agents: Git, GitHub Copilot, Claude Code, Kiro, Codex, NX Monorepo, Turborepo, Agile (Scrum/Jira), OAuth 2.0, Jest, Maestro (React Native E2E), NestJS Unit Integration Tests, Stripe, RBAC, LangChain, OpenAI/Ollama, Sentry, Vercel, CMS (Directus, Sanity, Payload), Microsoft Clarity, MCP